

Citation under Pressure: What Deployment Constraints Do to Legal Citation Integrity, and What Deterministic Verification Can Repair

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Abstract. Sanctions over fabricated legal citations have made language-model drafting a problem for courts, and most research on the problem has produced detection benchmarks that score errors in finished briefs. This paper measures how the errors are produced. Seven models from seven vendors, from a 30B open-weight model to frontier systems, drafted argument sections for 48 U.S. Supreme Court matters under five prompting conditions: baseline, a citation quota, a restriction to pre-1970 authority, a sanctions warning, and all three combined. Every citation in the 1,680 drafts was checked deterministically, with no language-model judge in any primary measurement: existence against an 18-million-row reporter index, quotations against a verbatim checker validated on seeded faults, and pinpoint pages against official reporter pagination from public archives. Under combined pressure the smallest model’s citation existence fell from 0.924 to 0.773, and verbatim quotation accuracy fell significantly for three of seven models; at the top of the range no contrast survived correction. In a second experiment the checker fed its findings back into revision. Sonnet 5 reached 0.974 verbatim accuracy, and a per-quotation trace gives the mechanism. Of its 89 flagged quotations, 4 were corrected; the rest were deleted or stripped of their quotation marks. False feedback produced the same deletions aimed at correct quotations, five of seven models ended less accurate than they started, and a verifier right half the time cost Sonnet 5 more than twice the correct quotations a precise one did. We release the instruments, including a page-level pinpoint verifier built from public data.

Keywords. legal citation, hallucination, large language models, verification, evaluation

1. Introduction

Since 2023, courts have repeatedly sanctioned lawyers who filed briefs containing citations that a language model invented, beginning with *Mata v. Avianca* [1] and continuing in a growing docket that a public tracker now counts in the hundreds [2]. Most published work frames the problem as detection: given a brief that already contains errors, how many can a model or agent find [3,4]? Detection accuracy is the right measure for re-

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viewing a filed brief; it does not tell a deployer which prompting conditions produce the errors, which is the question this paper addresses.

This paper asks two production-side questions. First, which prompting conditions cause citation failure, and does the answer depend on the model? Deployed prompts can carry constraints of the kind Zhao et al. [5] studied for scholarly citations, where a minimum count of authorities, a restriction on which may be cited, and a warning about consequences collapsed citation existence even in capable models; we are not aware of a comparable study for law. Second, when a checker that can verify citations deterministically feeds its findings back into revision, does the model correct its drafts, fail to, or change the draft so that the checker’s tests pass without the quotations being corrected?

Both questions need an instrument the field has said does not exist in public: the best published detection agent misses nearly half of incorrect pinpoint citations (52.8 percent recall; open-weight detectors 19 to 51 percent), and its authors conclude that “no models can reliably detect wrong pincites” because page-level information lives behind commercial databases [3]. We built it from public sources: existence and name pairing against a local index derived from CourtListener bulk data, verbatim quotation checking validated on seeded faults, and page-level pinpoint verification against the official reporter pagination in the Caselaw Access Project’s volume archives [6]. Every primary number reported here is produced by this instrument; no language-model judge is used in any primary measurement.

Our scope is the closed-book use of general-purpose models, the setting of most sanctioned filings documented to date [2,7]; Section 7 says what we expect to transfer to retrieval-grounded tools. The paper reports three results. First, pressure degrades citation integrity beginning with the smallest model, and the degradation is rarely accompanied by refusal; in the 1,344 drafts under the four non-baseline conditions (the pressured drafts) there were three refusals. Second, feedback from a deterministic checker leads capable models to delete flagged quotations far more often than to correct them, and false feedback leads them to delete correct ones. Third, at the top of the range the remaining errors are mostly genuine legal language attributed to the wrong case or the wrong page, which the released instruments detect.

2. Related Work

Legal citation reliability. Liu et al. [3] build a detection benchmark from real appellate briefs with injected errors and a taxonomy grounded in sanctioned filings; we adopt their error categories as outcome coding and measure the naturally occurring errors their limitations section asks for. LegalCiteBench [4] shows that models asked closed-book for the citation of a named case almost never produce the right one and almost never abstain; our task asks the easier question of whether the cases a model chooses to cite exist, and the low abstention transfers. Dahl et al. [7] measured hallucinated authority in general-purpose chatbots and Magesh et al. [8] in commercial legal research tools.

Constraints and fabrication. Zhao et al. [5] is the closest design: five prompting regimes over scholarly claims, verified by fuzzy bibliographic matching that left 36 to 61 percent of citations unresolved. We transplant their factorial to law with deterministic verification (0.2 percent unresolved) and a formal statistical model; their mechanism, compliance with the temporal window filled with fabrications, transfers to law for some models only (Section 4).

Revision with execution feedback. The loop protocol follows execution-feedback work in code generation [9], including a no-feedback control that separates feedback content from extra revision attempts; we add controls in which half or all of the feedback is false.

AI and law. Blair-Stanek and Van Durme [10] found at ICAIL that frontier models flip the winning party across identical zero-temperature runs; we ran each cell once, and Section 8 treats that instability as a limitation of this design. Evaluation methodology in this community has moved toward separating factual truth from reasoning soundness [11] and pairing automated metrics with the checks lawyers perform [12]; citation integrity is a check for which deterministic ground truth is available, and this paper automates it. Closest to our pinpoint instrument, Garneau et al. [13] found that retrieving the paragraph a proposition rests on is far harder than retrieving the case; our verifier approaches the same gap from the other side, checking a generated pincite against the page text. Zhang et al. [14] show models miss overruled precedent, a failure adjacent to the wrong-authority errors we measure, and ConsRAG [15] constrains retrieval to prevent hallucination, the strategy complementary to the post-hoc feedback our loop tests. On trust in model-drafted documents, Trozze et al. [16] found mock jurors’ decisions unrelated to whether a complaint was written by ChatGPT or a lawyer, and Harasta et al. [17] found lawyers rate identical documents lower when told a model wrote them; machine verification of citation integrity offers a basis for trust that depends on neither perception. Gray et al. [18] and Pires et al. [19] evaluate generated legal argument and legal writing without checking the authority the text cites, the layer this paper adds; citation-worthiness detection [20] asks where a citation is needed, and the citation index behind our checker is the kind of network studied in [21,22,23], used here to test whether a cited case exists.

3. Tasks, Conditions, and Instruments

Terms. In a citation such as *Brown v. Board of Education*, 347 U.S. 483, 495 (1954), the volume, reporter, and first page identify the case and the optional second page (495) is the *pincite*, which for a quotation must be the page where the quoted language appears. Throughout, *citation existence* is the share of adjudicable citations that resolve to a real case, and *strict quotation accuracy* is the share of scored quotations that reproduce the opinion verbatim.

Matters. The matters are forty-eight Supreme Court cases from 1990 to 2020, drawn from a 500-case pool built from the Supreme Court Database [24]. Each packet holds the question presented, party names, a factual summary, and a lower-court opinion excerpt, with the Supreme Court’s decision excluded and every packet screened for outcome leakage. The model argues an assigned side (petitioner for even-numbered matters, respondent for odd) in an argument section of under 900 words; nothing in the design predicts outcomes, so the memorization confound of Supreme Court prediction tasks [25] is out of scope.

Conditions. The five conditions are baseline; quota (“cite at least eight distinct case authorities”); temporal (“every authority you cite MUST have been decided before 1970”); stakes (“this section will be filed with the court; citing authority that does not exist, or misquoting authority, risks sanctions under Rule 11”); and combined, which con-

catenates the three clauses. Each condition has one fixed template, run at temperature 0. The stakes clause names Rule 11 of the Federal Rules of Civil Procedure because it is the sanctions rule invoked in the sanctioned-filing cases [1]; Rule 11 governs civil actions in the district courts [26], so the clause serves only as a stakes signal for a Supreme Court filing. Each condition is a single clause so that its effect can be isolated; a robustness check re-ran twelve matters under alternative phrasings of every condition.

Models. The models are Qwen3-30B-A3B, Mistral Small, Llama-4 Maverick, DeepSeek V4 Flash, Grok 4.3, GPT-5.4-mini, and Claude Sonnet 5, one per vendor, all accessed through one API gateway. The set spans open-weight 30B to frontier with no vendor contributing two models; we do not assume a total ordering, tables are ordered by baseline strict quotation accuracy, and Sonnet 5, GPT-5.4-mini, and Grok 4.3, with the highest baseline existence, are called the top of the range. The gateway identifiers, run through OpenRouter in August 2026 with completion limits of 1,400 to 12,000 tokens, are qwen3-30b-a3b-instruct-2507, mistral-small-2603, llama-4-maverick, deepseek-v4-flash, grok-4.3, gpt-5.4-mini, and claude-sonnet-5. An eighth model, GLM-4.7-flash, completed 207 of 240 drafts and is reported in Section 4.

Adjudication. Citations are extracted with eyecite [27] and given one of three labels: exists, not found, or unverifiable (vendor citations and coverage gaps), so that index coverage is never counted as fabrication; 0.2 percent of the 14,040 citation records in the full run are unverifiable. Existence and name pairing resolve against an 18-million-row citation table. Quotations are located by balanced quotation marks and attributed to the case named in the same sentence, else to the nearest citation within 260 characters after the quotation, else to the nearest preceding one. Each is checked verbatim against the cited opinion’s text from public sources; a quotation is *near miss* when every fragment reaches token coverage of at least 0.85 without matching exactly, and *inaccurate* otherwise. The strict rate for a draft d is $S_d = n_{\text{acc}} / (n_{\text{acc}} + n_{\text{near}} + n_{\text{inacc}})$, and a lenient rate counting near misses as correct is reported alongside it. Before any measurement the quotation checker scored 100 of 100 on twenty synthetic drafts with planted genuine quotations, wrong attributions, single-word corruptions, fabrications, and formatting artifacts; Section 6 also audits its verdicts on real drafts. Pinpoint pages are verified against the official reporter pagination parsed from the Caselaw Access Project’s HTML volumes, which mark every page break; a quotation counts as on the cited page when its longest fragment lies within that page’s text, and adjacent-page matches are reported separately.

Human reference. The same instrument scored 482 clean, pre-ChatGPT federal appellate brief excerpts from Liu et al. [3]; human lawyers reach 0.407 strict and 0.680 lenient quotation accuracy (578 scored quotations) and 0.931 citation existence, the shortfall being index coverage of state and unpublished reporters. Human briefs cite more non-federal authority and quote statutes and records more often than the models, so the human figures are a reference for the severity of the verbatim standard only.

Inference. The primary analysis is a citation-level logistic generalized estimating equation per model, $\text{logit} P(y_{imk} = 1) = \beta_0 + \beta_k$ for citation i in matter m under condition k , with exchangeable working correlation within matters [28], so that the 48 matters rather than the citations are the independent units. Each model has 8 contrasts (four conditions, two outcomes); we report Holm-corrected p values within model and call an effect significant only when it survives correction, so statements that count models (“three of seven”) are descriptive. Odds ratios, 95 percent intervals, and raw and corrected

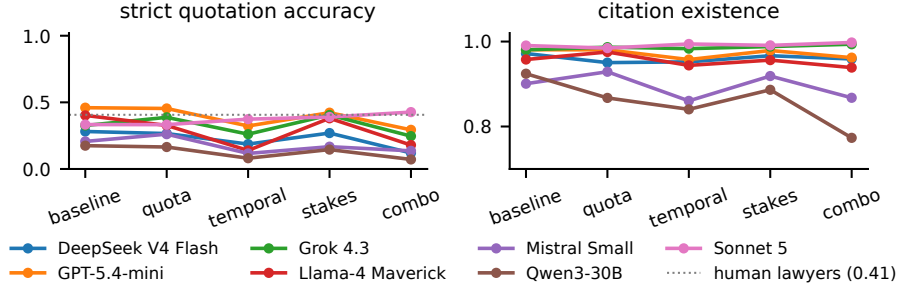


Figure 1. Strict quotation accuracy (left) and citation existence (right) by condition, one line per model. The dotted line is the human-lawyer reference under the identical instrument. The right panel’s axis starts at 0.7.

Table 1. Baseline versus the combined condition, rows ordered by baseline strict quotation accuracy. Quotation columns give the strict rate with the lenient rate in parentheses; existence columns give the rate with the raw count of citations not found in brackets. An asterisk marks a contrast significant after Holm correction within model.

Model	Quotes, base	Quotes, comb.	Exist, base	Exist, comb.	Temporal viol.
Qwen3-30B	0.175 (0.564)	0.072 (0.322)*	0.924 [28]	0.773 [90]*	20%
Mistral Small	0.207 (0.608)	0.136 (0.383)	0.901 [37]	0.868 [69]	35%
DeepSeek V4 Flash	0.281 (0.631)	0.120 (0.439)*	0.972 [10]	0.959 [20]	11%
Grok 4.3	0.326 (0.644)	0.246 (0.578)	0.981 [5]	0.994 [3]	0%
Sonnet 5	0.333 (0.622)	0.427 (0.642)	0.991 [3]	0.998 [1]	0%
Llama-4 Maverick	0.402 (0.618)	0.179 (0.453)*	0.958 [9]	0.939 [31]	10%
GPT-5.4-mini	0.460 (0.722)	0.294 (0.571)	0.981 [8]	0.962 [27]	13%

p values for all 56 contrasts are in the released results, and a matter-level cluster bootstrap over draft rates agreed in sign with every corrected result. For the revision experiment, which was pre-registered descriptively, we add matter-paired comparisons of final strict accuracy between arms (Wilcoxon signed-rank test and a bootstrap interval on the mean difference), chosen after the data were seen. Hypotheses and decision rules were frozen before the full run; every later analysis decision, including two withdrawn findings, is recorded in a dated ledger released with the code.

4. Experiment 1: Pressure

Table 1 and Figure 1 report the rates.

Citation existence. Only the 30B model’s existence moves under pressure: from 0.924 at baseline to 0.773 under the combined condition, with the quota, temporal, and combined contrasts all significant after correction (combined OR 0.30, 95 percent CI 0.18 to 0.51, corrected $p < 0.001$). Mistral Small falls under the temporal and combined conditions without reaching significance, and the five other models stay between 0.93 and 1.00 in every condition.

Quotation accuracy. The temporal restriction is the condition that damages quoting. After correction, strict quotation accuracy falls significantly under temporal and com-

bined pressure for Qwen3-30B (0.175 to 0.072) and Llama-4 (0.402 to 0.179), and under the combined condition for DeepSeek (0.281 to 0.120); GPT-5.4-mini, Mistral Small, and Grok 4.3 fall in the same direction without surviving correction, and Sonnet 5 does not fall at all. The Llama-4 row shows the largest drop from a baseline at the human level: at baseline it quotes at the human reference level, 0.402 against 0.407, and under the temporal restriction its accuracy drops to 0.139 (OR 0.20, CI 0.10 to 0.40, corrected $p < 0.001$). The lenient rates in Table 1 move with the strict rates for every model, so the result does not depend on how bracketed alterations are scored.

A pre-registered contrast that did not survive correction. Sonnet 5’s strict quotation accuracy is numerically highest under the combined condition (0.427 against 0.333; uncorrected $p = 0.0294$, corrected $p = 0.235$). The stakes condition alone moves nothing, and the temporal restriction shifts the cited cases toward older, better-memorized opinions (median decision year 1943 against 1986 at baseline), an untested alternative to any reading in terms of more conservative citing. We report it as a null result.

Compliance with the temporal rule. Violations of the pre-1970 restriction (Table 1) run from 35 percent of adjudicable citations for Mistral Small to 1 in 803 for Sonnet 5 and 2 in 719 for Grok 4.3. Mistral Small violates the rule most while its existence and quotation figures are second-lowest, so compliance does not track the other outcomes in rank.

Refusals and one model that stalls. Across the 1,344 pressured drafts there were three refusals (two from DeepSeek, one from GPT-5.4-mini). GLM-4.7-flash produced no output on 33 of its 240 drafts, 28 of them in the combined condition, reasoning until a 30,000-token budget was exhausted; it is excluded from Table 1 and reported as a third failure mode beside fabrication and refusal.

5. Experiment 2: The Checker inside the Revision Loop

Combined-condition drafts entered a revision loop. Each round, the checker returned one line per failure (a citation that does not resolve; a quotation that does not appear verbatim in the cited opinion), the model revised, and the loop ended on a clean draft or after three revisions. A control with *false feedback* gave the same number of identically formatted lines pointed at items that were correct; a *half-true* control kept each line or replaced it with a false one with equal probability, so the verifier’s precision was about one half; and a *no-feedback* control gave the same number of revision rounds with a generic instruction to revise and no list of failures. The loop used the first 24 matters in the pre-registered order (twelve per side); loop rates are means over drafts on 24 matters, so round-0 figures differ from the pooled citation-level rates over 48 matters in Table 1, and citation counts in this section include unverifiable citations. Every scored quotation in a round-0 draft was traced to the final draft as *corrected* (the same or an edited quotation on the same citation is now accurate), *dequoted* (its words remain without quotation marks), or *deleted*.

Satisfying the checker by deletion. Under true feedback the final strict rate ranges from 0.107 for Qwen3-30B to 0.974 for Sonnet 5 (Table 2), and the rate alone is consistent with repair; the trace shows that the rise comes from deletion. Sonnet 5 had 89 flagged quotations at round 0: 4 were corrected, 13 kept their words and lost their quotation marks, and 72 were deleted; 17 of its 68 accurate quotations were removed as well.

Table 2. Revision under true checker feedback, 24 episodes per model. Round-0 quotations, flagged (inaccurate or near miss) and accurate, traced to the final draft; strict accuracy round 0 to final; Clean counts episodes ending on a draft the checker passed.

Model	n	Flagged at round 0			Accurate at round 0		Strict	Clean
		corrected	dequoted	deleted	n	removed		
Qwen3-30B	170	0	11	30	16	0	0.099 to 0.107	0
Mistral Small	130	0	12	39	15	2	0.208 to 0.264	9
DeepSeek V4 Flash	183	0	67	101	28	12	0.181 to 0.645	21
Grok 4.3	46	0	27	19	15	5	0.207 to 1.000	24
Sonnet 5	89	4	13	72	68	17	0.471 to 0.974	23
Llama-4 Maverick	53	0	9	28	11	5	0.176 to 0.349	21
GPT-5.4-mini	37	2	5	21	28	2	0.360 to 0.719	22

Table 3. Verifier precision and its cost. The four arms differ only in the share of feedback lines that are true: all (true), half, none (false), or no feedback at all. Left: final strict quotation accuracy (round 0 is the same draft in every arm). Right: accurate round-0 quotations that the revision deleted, dequoted, or made inaccurate, of the n in Table 2.

Model	round 0	Strict accuracy, final				Accurate quotations removed			
		true	half	false	none	true	half	false	none
Qwen3-30B	0.099	0.107	0.097	0.093	0.072	0	2	4	2
Mistral Small	0.208	0.264	0.223	0.122	0.198	2	7	8	2
DeepSeek V4 Flash	0.181	0.645	0.325	0.153	0.152	12	21	24	11
Grok 4.3	0.207	1.000	0.667	0.208	0.236	5	9	12	7
Sonnet 5	0.471	0.974	0.664	0.266	0.595	17	38	56	18
Llama-4 Maverick	0.176	0.349	0.400	0.188	0.109	5	10	6	8
GPT-5.4-mini	0.360	0.719	0.706	0.241	0.543	2	7	15	9

Its scored quotations fell from 157 to 73 while its accurate ones went from 68 to 70 (15 newly added). Grok 4.3, which quotes sparingly (61 scored quotations at round 0), reached 1.000 with every episode clean by correcting none of its 46 flagged quotations and dequoting or deleting all of them. Citation existence, by contrast, was repaired: the loop existence rate rose from 0.739 to 0.822 for Qwen3-30B and from 0.930 to 1.000 for Llama-4 while citation counts fell by at most a tenth, so unresolvable citations were replaced. At every capability tested, checker feedback led models to remove flagged quotations or convert them to unmarked paraphrase far more often than to correct them; neither outcome is visible to the checker or to a reader of the revised draft.

Three controls: feedback content and verifier precision. Table 3 and Figure 2 separate feedback content from revision itself and vary the verifier’s precision. Without feedback, Sonnet 5’s strict rate still rose (0.471 to 0.595), and true feedback added 0.36 over revision alone (matter-paired mean difference; bootstrap interval 0.23 to 0.51, Wilcoxon $p=0.0003$); the same holds for DeepSeek (0.49, $p=0.005$) and Llama-4 (0.29, $p=0.039$). For GPT-5.4-mini the difference between true feedback and no feedback is 0.12 with an interval that includes zero, so its apparent repair is not distinguishable from what revision alone does; Qwen3-30B moves little in any arm, and Mistral Small gains little from true feedback but falls to 0.122 under false feedback (true minus false 0.12, $p=0.012$). The right half of the table gives the cost of imprecision. Reading across the

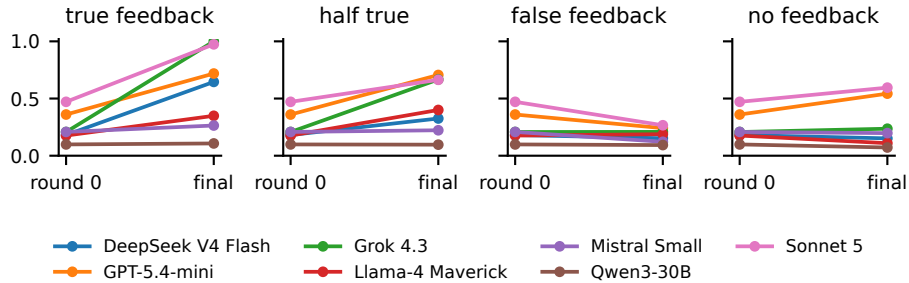


Figure 2. Strict quotation accuracy from round 0 to the final draft under true, half-true, false, and no feedback, one color per model. A rate can rise because accurate quotations were added or because inaccurate ones were removed; Tables 2 and 3 separate the two.

Sonnet 5 row, it removed 17 of its 68 accurate quotations when every flag was true, 38 when half were, and 56 when none were, against 18 with no feedback. A verifier that is right half the time left Sonnet 5 at 0.664, below true feedback by 0.31 ($p=0.001$) and within seven points of revision with no feedback at all (0.595). Under all-false feedback five of seven models ended below their starting rate (Llama-4 and Grok 4.3 moved by about a point), and inaccurate counts fell in every model, so what changed was the deletion of correct quotations. Our checker’s own false-alarm rate, measured by the provenance audit of Section 6, was 1.2 percent of Sonnet 5’s inaccurate verdicts (9 of 746).

6. Residual Errors at the Top of the Range

Three error classes remain when fabrication is rare, each measured deterministically.

Paraphrase inside quotation marks. Decomposing every inaccurate quotation against the cited opinion’s own text, 55 percent of Sonnet 5’s inaccurate quotations (413 of them) are recognizable paraphrases of the correct case presented as verbatim quotation, 202 match nothing substantial in the cited opinion, and 131 cite authority outside the text store’s scope; GPT-5.4-mini’s split is 52 percent paraphrase.

Real language, wrong case. Searching every inaccurate quotation against a segmented corpus of 8,392 Supreme Court opinions locates 626 quotations across the seven models (127 for Sonnet 5, 77 for Qwen3-30B) whose language appears verbatim in a different opinion than the one cited (matched probes are at least six words, so stock legal phrases do not count); quotations found nowhere in the corpus (511 for Sonnet 5) include the paraphrases of the preceding paragraph and quotations from other courts. The same search audits the checker: it classified 9 of Sonnet 5’s inaccurate verdicts and 9 of Qwen3-30B’s as the checker’s own error, the language being present in the attributed case. A screen for models reproducing the argued case’s own later opinion from memory found 41 candidates, all record text from the prompt or doctrinal formulas the opinions were themselves quoting; the finding was withdrawn.

Right case, wrong page. Where a pinpoint and a quotation co-occur on a U.S. Reports authority, the quotation lies on the cited page 77 percent of the time for Sonnet 5, between 42 and 64 percent for the other six, against 89 percent for the human briefs of Section 3 (45 quotations), and on a different page of the cited case 16 to 45 percent of

the time (adjacent-page matches make up the remainder); pins outside the cited case are rare at every tier.

Relevance judging: three failed calibrations. Whether a real, correctly quoted case supports the proposition has no deterministic test. Three language-model panels calibrated against the expert-labeled content-misrepresentation items of Liu et al. [3], with a pre-declared bar of 0.75 accuracy, scored 0.500, 0.700, and 0.594 (misrepresentation recall at best 4 of 17); no verdict from a failed panel was interpreted, and the result joins evidence that agreement statistics overstate judge validity [29,30].

7. Discussion

Where the checker belongs. The revision experiment argues for placing the checker at the gate, where it holds a draft and returns the failures to a person, and against placing it in the loop, where the model is asked to resolve them. The two positions use the same instrument and differ only in who acts on its output. Table 2 shows what Sonnet 5 did when it acted: it resolved 89 flagged quotations by correcting 4 and removing the rest. A drafting tool built on the loop pattern will report a clean draft, and the supervising attorney will not learn that claims once presented as quotations are now paraphrase; with the checker at the gate the same list of failures reaches the attorney (20 of the 168 combined-condition drafts entering the loop would have passed unrevised).

Verifier precision is a deployment requirement. The false-feedback arm measures what an imprecise verifier costs. Our checker’s own false-alarm rate on Sonnet 5’s drafts was 1.2 percent. Published detection agents are less precise: Liu et al. [3] report that weaker detectors flag a substantial share of correct citations, reading a coverage gap in a public index as a fabrication. Table 3 shows what such flags do once a capable model acts on them, and the half-true arm shows the cost rising with the share of false flags. At a precision of one half, Sonnet 5 lost more than twice the correct quotations it lost under a precise verifier and ended within seven points of where revision without feedback left it. A pipeline that feeds a probabilistic detector’s flags to a model should therefore report the detector’s precision alongside its recall, since precision sets how many correct quotations the model will remove. For existence, verbatim quotation, and pinpoint page no probabilistic detector is needed, because the check is exact and the released instrument performs it from public data.

What evaluations should include. A single fabrication rate at a neutral prompt misses both effects reported here: the 30B model’s baseline existence of 0.924 overstates what it delivers under the constraints a user is likely to add, and quotation accuracy, which existence benchmarks omit, fell for six of the seven models. The temporal restriction was the most damaging single clause and the one to include first, since a date or jurisdiction restriction is the kind of constraint a drafting prompt can carry. The residual classes at the top of the range, paraphrase inside quotation marks and quotations on the wrong page of the right case, are invisible to existence checks, and we expect retrieval-grounded tools to leave them in place, since they arise when the model composes the quotation.

For courts. Court orders that require disclosure of AI assistance [2] ask filers to state whether a model was used; the three checks reported here can be run by the court itself, from public data, on any filing, and cover the failures named in the sanctions

decisions [1]. What remains for human judgment is whether a real, correctly quoted case supports the proposition, and Section 6 shows that language-model panels are not yet a substitute.

8. Limitations

The study covers one domain and task, Supreme Court merits argument in one packet format with one fixed template per condition, closed-book, and the stakes clause names a sanctions rule that does not govern the forum. The no-feedback and half-true controls and the per-quotation trace were added after the pre-registered analysis, and the tests in Section 5 were chosen after the data were seen. Each cell was generated once at temperature 0. A second, independent generation of the baseline and combined drafts for twelve matters on the six original models produced 1 draft identical to the first among 144, the instability reported by Blair-Stanek and Van Durme [10]; existence rates agreed within 0.031 and the Qwen3-30B existence contrast, the only one significant in the full run, kept its sign (it flipped for DeepSeek and GPT-5.4-mini, whose differences were within 0.05 of zero), while strict quotation rates on twelve matters differed by a median of 0.061 and up to 0.328 (GPT-5.4-mini, combined), where the contrast changed sign. Per-cell quotation rates therefore carry run-to-run noise that 48 matters reduce but do not remove. On a twelve-matter check under alternative phrasings of every condition (four models), combined-condition existence stayed within 0.06 of the original phrasing and no existence result changed sign, but strict quotation rates moved by up to forty points in either direction, so the quotation rates in Table 1 are specific to the templates used; the claims we rely on are the within-model contrasts that survive correction. Inference is per model at 48 matters, so effects of a few points are not detectable, and the seven models are one per vendor rather than a sampled population, so capability, family, and training recipe are confounded.

9. Conclusion

Under constraints of the kind found in practice, citation integrity degrades from the smallest model upward: nonexistent citations at 30B, misquotation through the middle of the range, and misattribution and wrong pages at the top, with three refusals in 1,344 drafts. A checker built from public data detects each class, including the page-level errors thought to require commercial databases. Returned to the model, its findings repair citation existence; flagged quotations are removed far more often than corrected, and false flags remove correct ones. We therefore recommend placing verification at the gate and reporting verifier precision for any system that acts on its output.

10. Ethics and Release

All source material is public court data; no generated draft was filed. <https://github.com/hardy30894/citation-under-pressure> holds the instruments, drafts, records, and analysis ledger, so the same checks can be run on any brief, by courts as well as by drafters. Language-model assistance was used in writing code and prose under the author’s direction; every reported number comes from the released instruments.

References

- [1] *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443 (S.D.N.Y. 2023); 2023. Opinion and order on sanctions, No. 1:22-cv-01461 (PKC), June 22, 2023.
- [2] Charlotin D. AI Hallucination Cases Database; 2025. Accessed 2 September 2026; 2,008 cases listed. <https://www.damiencharlotin.com/hallucinations/>.
- [3] Liu P, Stambach D, Henderson P. Who Checks the Citations? Benchmarking Legal Hallucination Detection; 2026. Accepted at the Conference on Language Modeling (COLM) 2026. Available from: <https://arxiv.org/abs/2606.21155>. doi:10.48550/arXiv.2606.21155.
- [4] Chen S, Yin H, Zhou S. LegalCiteBench: Evaluating Citation Reliability in Legal Language Models; 2026. Available from: <https://arxiv.org/abs/2605.10186>. doi:10.48550/arXiv.2605.10186.
- [5] Zhao C, Tang Y, Qian Y. Do Deployment Constraints Make LLMs Hallucinate Citations? An Empirical Study across Four Models and Five Prompting Regimes; 2026. Available from: <https://arxiv.org/abs/2603.07287>. doi:10.48550/arXiv.2603.07287.
- [6] Harvard Law School Library Innovation Lab. Caselaw Access Project; 2024. Digitized collection of official, book-published U.S. case law; accessed September 2026. <https://case.law/>.
- [7] Dahl M, Magesh V, Suzgun M, Ho DE. Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. *Journal of Legal Analysis*. 2024;16(1):64-93. doi:10.1093/jla/laae003.
- [8] Magesh V, Surani F, Dahl M, Suzgun M, Manning CD, Ho DE. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. *Journal of Empirical Legal Studies*. 2025;22(2):216-42. doi:10.1111/jels.12413.
- [9] Gehring J, Zheng K, Copet J, Mella V, Cohen T, Synnaeve G. RLEF: Grounding Code LLMs in Execution Feedback with Reinforcement Learning. In: *Proceedings of the 42nd International Conference on Machine Learning*. vol. 267 of *Proceedings of Machine Learning Research*. PMLR; 2025. p. 19034-55. Available from: <https://proceedings.mlr.press/v267/gehring25a.html>.
- [10] Blair-Stanek A, Van Durme B. LLMs Provide Unstable Answers to Legal Questions. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. Chicago, IL, USA: ACM; 2025. p. 425-9. doi:10.1145/3769126.3769245.
- [11] Gultekin S, Rossi Reich M, Galloni F, Lagioia F, Consiglio E, Sartor G, et al. A Two-Dimensional Evaluation Framework for Factual and Reasoning Assessment of LLMs in Legal Question Answering. In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 86-97. doi:10.3233/FAIA251579.
- [12] Sanchi M, Novotná T. Towards Hybrid Evaluation Methodologies for Large Language Models in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 389-92. doi:10.3233/FAIA241279.
- [13] Garneau N, Olsen HP, Corduant A, Tarissan F. Combining Network and Text to Provide Legal Pincites. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 282-7. doi:10.3233/FAIA241256.
- [14] Zhang L, Savelka J, Ashley KD. Do LLMs Truly “Understand” when a Precedent Is Overruled? In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 239-50. doi:10.3233/FAIA251592.
- [15] Nguyen HT, Satoh K. ConsRAG: Minimize LLM Hallucinations in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 327-32. doi:10.3233/FAIA241263.
- [16] Trozze A, Davies T, Kleinberg B. Large language models in cryptocurrency securities cases: can a GPT model meaningfully assist lawyers? *Artificial Intelligence and Law*. 2025;33(3):691-737. doi:10.1007/s10506-024-09399-6.
- [17] Harasta J, Novotná T, Savelka J. It cannot be right if it was written by AI: on lawyers’ preferences of documents perceived as authored by an LLM vs a human. *Artificial Intelligence and Law*. 2026;34(1):153-90. doi:10.1007/s10506-024-09422-w.
- [18] Gray MA, Savelka J, Oliver WM, Ashley KD. Generating Legal Arguments with Automatically Identified Factor Magnitudes. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 199-208. doi:10.1145/3769126.3769212.
- [19] Pires R, Malaquias Junior R, Nogueira R. Automatic Legal Writing Evaluation of LLMs. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 420-4. doi:10.1145/3769126.3769227.

- [20] Khatri M, Sheik R, Wadhwa P, Satija G, Kumar Y, Shah RR, et al. CiteCaseLAW: Citation Worthiness Detection in Caselaw for Legal Assistive Writing. In: Legal Knowledge and Information Systems: JURIX 2023, vol. 379 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2023. p. 257-62. doi:10.3233/FAIA230971.
- [21] Sadeghian A, Sundaram L, Wang DZ, Hamilton WF, Branting K, Pfeifer C. Automatic semantic edge labeling over legal citation graphs. Artificial Intelligence and Law. 2018;26(2):127-44. doi:10.1007/s10506-018-9217-1.
- [22] Dadgostari F, Guim M, Beling PA, Livermore MA, Rockmore DN. Modeling law search as prediction. Artificial Intelligence and Law. 2021;29(1):3-34. doi:10.1007/s10506-020-09261-5.
- [23] Mumford J, Florimonte F, Atkinson K, Dzehtsiarou K. Context-Aware Citation Networks: A Human-AI Dataset, Analysis, and Tool. In: Legal Knowledge and Information Systems: JURIX 2025, vol. 416 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2025. p. 145-55. doi:10.3233/FAIA251584.
- [24] Spaeth HJ, Epstein L, Nelson MJ, Martin AD, Segal JA, Ruger TJ, et al.. Supreme Court Database, Version 2025 Release 01; 2025. Modern database, terms 1946 to 2024. <https://scdb.la.psu.edu/>.
- [25] Ortega JE, Joshi DD, Borkowski MP. Large-Language Memorization During the Classification of United States Supreme Court Cases; 2025. Available from: <https://arxiv.org/abs/2512.13654>. doi:10.48550/arXiv.2512.13654.
- [26] Federal Rules of Civil Procedure, Rule 1 (Scope and Purpose); 2024. "These rules govern the procedure in all civil actions and proceedings in the United States district courts". https://www.law.cornell.edu/rules/frcp/rule_1.
- [27] Cushman J, Dahl M, Lissner M. eyecite: A tool for parsing legal citations. Journal of Open Source Software. 2021;6(66):3617. Available from: <https://doi.org/10.21105/joss.03617>. doi:10.21105/joss.03617.
- [28] Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. Biometrika. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13.
- [29] Norman JD, Rivera MU, Hughes DA. Reliability without Validity: A Systematic, Large-Scale Evaluation of LLM-as-a-Judge Models Across Agreement, Consistency, and Bias; 2026. Available from: <https://arxiv.org/abs/2606.19544>. doi:10.48550/arXiv.2606.19544.
- [30] Zheng L, Chiang WL, Sheng Y, Zhuang S, Wu Z, Zhuang Y, et al.. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena; 2023. NeurIPS 2023 Datasets and Benchmarks Track. Available from: <https://arxiv.org/abs/2306.05685>. doi:10.48550/arXiv.2306.05685.